

NORTH SYDNEY GIRLS HIGH SCHOOL

Section I

5 marks

Attempt Questions 1-5

Use the multiple-choice answer sheet for Questions 1-5

1 Which expression is equal to $\int \sin^2 5x \, dx$?

(A) $\frac{1}{2} \left(x - \frac{1}{5} \sin 5x \right) + C$

(B) $\frac{1}{2} \left(x + \frac{1}{5} \sin 5x \right) + C$

(C) $\frac{1}{2} \left(x - \frac{1}{10} \sin 10x \right) + C$

(D) $\frac{1}{2} \left(x + \frac{1}{10} \sin 10x \right) + C$

2 What is the domain and range of $y = \frac{3}{2} \cos^{-1} 2x$?

(A) $-2 \leq x \leq 2, \quad 0 \leq y \leq \frac{3\pi}{2}$

(B) $-\frac{1}{2} \leq x \leq \frac{1}{2}, \quad 0 \leq y \leq \frac{3\pi}{2}$

(C) $-2 \leq x \leq 2, \quad 0 \leq y \leq \frac{2\pi}{3}$

(D) $-\frac{1}{2} \leq x \leq \frac{1}{2}, \quad 0 \leq y \leq \frac{2\pi}{3}$

3 What is the derivative of $\tan^{-1} 4x$?

(A) $\frac{4}{1+16x^2}$

(B) $\frac{4}{16+x^2}$

(C) $\frac{2}{1+4x^2}$

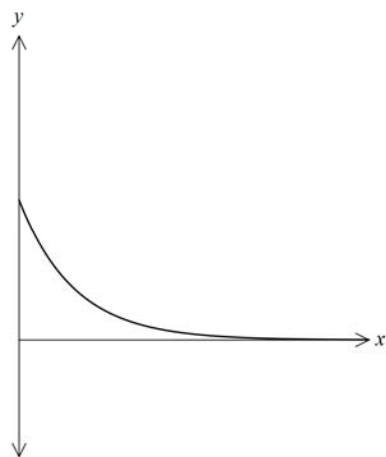
(D) $\frac{2}{4+x^2}$

4 What is the exact value of $\tan^{-1}\left(\tan \frac{5\pi}{6}\right)$?

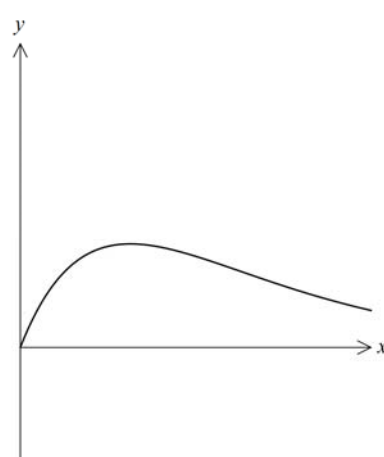
- (A) $-\frac{1}{\sqrt{3}}$ (B) $\sqrt{3}$ (C) $\frac{5\pi}{6}$ (D) $-\frac{\pi}{6}$

5 Which of the following best represents the graph of $y = xe^{-0.5x}$ for $x \geq 0$?

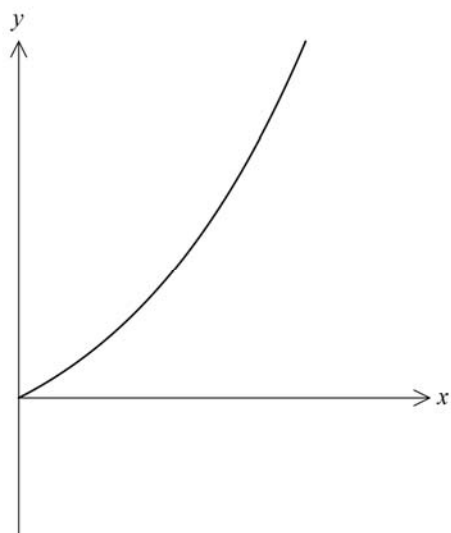
(A)



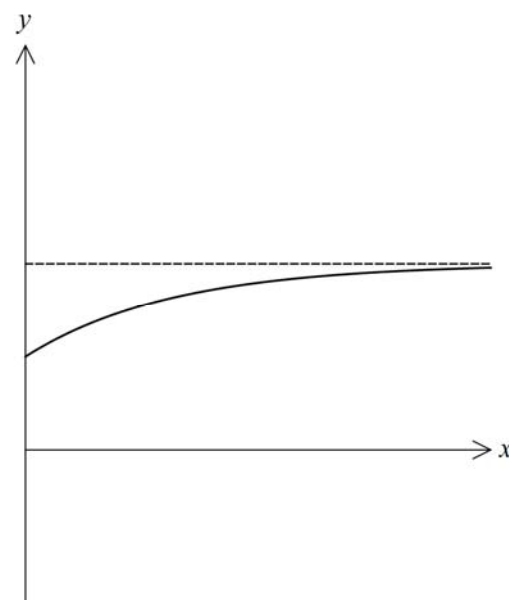
(B)



(C)



(D)



Section II

36 marks

Attempt Questions 6-8

Answer each question in a SEPARATE writing booklet. Extra writing paper is available.

In Questions 6-8, your responses should include relevant mathematical reasoning and/or calculations.

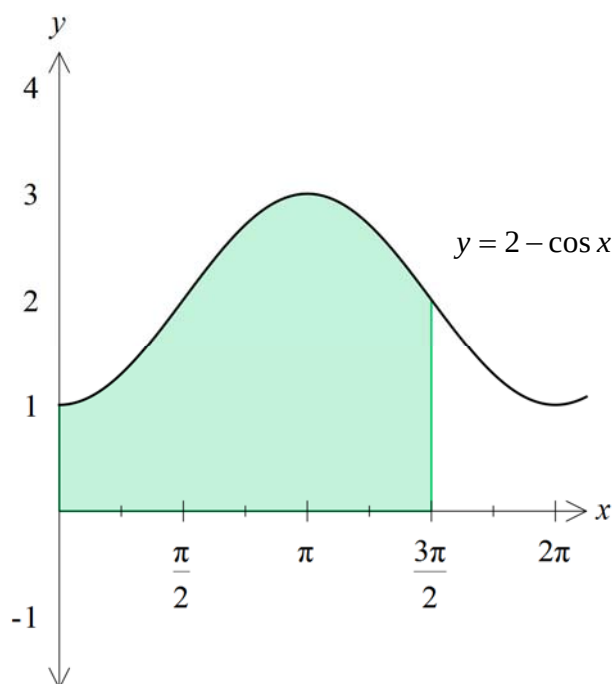
Question 6 (12 marks)

- (a) Find the primitive of $\frac{1}{9+4x^2}$ 2
- (b) Find the exact value of $\int_0^{\frac{1}{2}} \frac{x}{1+4x^2} dx$. 2
- (c) Sketch the function $y = 2\sin^{-1}(x+1)$ 2
- (d) Use the substitution $u = e^x$ to find $\int_{-\ln 2}^0 \frac{e^x}{\sqrt{1-e^{2x}}} dx$ 3
- (e) Write a general solution of the equation $\tan x = -\sqrt{3}$ 2
- (f) Evaluate $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{2\theta}$ 1

Question 7 (12 marks) Use a SEPARATE writing booklet.

- (a) The region bounded by the axes and the curve $y = 2 - \cos x$ between $x = 0$ and $x = \frac{3\pi}{2}$ is rotated about the x axis. Find the exact volume of the solid of revolution.

3



- (b) (i) Find the equations of the horizontal and vertical asymptotes of the function

2

$$f(x) = \frac{1}{\sqrt{4x+3}}.$$

- (ii) Find the derivative of the function. 1
- (iii) On a one-third page diagram, sketch $y = f(x)$ and the line $y = x$ using the same scale on both axes. 2
- (iv) On the same diagram, sketch the inverse function $y = f^{-1}(x)$. 2
- (v) Find the equation of $y = f^{-1}(x)$, noting any restrictions on the domain. 2

Question 8 (12 marks) Use a SEPARATE writing booklet.

(a) Find the equation of the tangent to the graph $y = \sin^{-1}(2x-1)$ at $x = \frac{1}{4}$. 3

(b) (i) Show that the derivative of $f(x) = 2(\sqrt{x}-1) - \log_e x$, $x > 0$ is given by 2

$$f'(x) = \frac{\sqrt{x}-1}{x}.$$

(ii) Find the coordinates and the nature of the stationary point on the curve. 2

(iii) Evaluate $\lim_{x \rightarrow \infty} f'(x)$. 1

(iv) Sketch the curve, showing the features found in parts (ii) and (iii). 2

(v) Explain why $\frac{\log_e x}{x} < \frac{2}{x}(\sqrt{x}-1)$ for $x > 1$. 2

End of test

Multiple Choice

Ext 1 - HSC Task 3 June 2014

$$1) \int \sin^2 5x \, dx = \frac{1}{2} \int (1 - \cos 10x) \, dx$$

$$= \frac{1}{2} \left(x - \frac{1}{10} \sin 10x \right) + C \quad (C)$$

$$2) y = \frac{3}{2} \cos^{-1} 2x$$

$$\text{Domain: } -\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$\text{Range: } 0 \leq y \leq \frac{3\pi}{2} \quad (B)$$

$$3) \frac{d}{dx} [\tan^{-1}(4x)] = \frac{1}{1+(4x)^2} \times 4$$

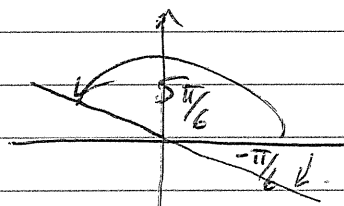
$$= \frac{4}{1+16x^2} \quad (A)$$

$$4) \tan^{-1} \left[\tan \left(\frac{5\pi}{6} \right) \right]$$

$$= \tan^{-1} \left[\tan \left(-\frac{\pi}{6} \right) \right]$$

$$= -\frac{\pi}{6}$$

(D)



$$5) y = xe^{-0.5x} \text{ for } x \geq 0$$

$$x=0, y=0$$

$$y' = x(-0.5e^{-0.5x}) + e^{-0.5x}$$

$$= e^{-0.5x} \left(1 - \frac{x}{2} \right)$$

$$\therefore y' = 0 \text{ when } x = 2 \Rightarrow (B)$$

Also, as $x \rightarrow \infty$, $e^{-0.5x} \rightarrow 0$ and it dominates $y = x$

$$\therefore xe^{-0.5x} \rightarrow 0 \text{ as } x \rightarrow \infty.$$

In Summary 1) C 2) B 3) A 4) D 5) B

Question 6

$$\begin{aligned} \text{a) } \int \frac{1}{9+4x^2} dx &= \frac{1}{4} \int \frac{1}{\frac{9}{4}+x^2} dx \quad a^2 = \frac{9}{4} \\ &= \frac{1}{4} \times \frac{2}{3} \int \frac{3/2}{\frac{9}{4}+x^2} dx \\ &= \frac{1}{6} \tan^{-1} \frac{2x}{3} \end{aligned}$$

Alternatively:

$$f(x) = 2x$$

$$a^2 = 3$$

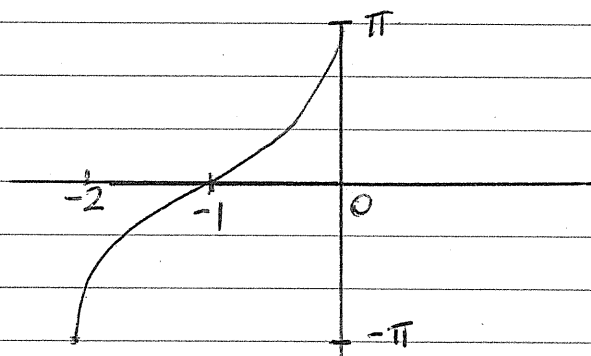
$$\int \frac{1}{a^2 + [f(x)]^2} = \frac{1}{a} \times \frac{1}{f'(x)} \tan^{-1} \frac{f(x)}{a}$$

$$= \frac{1}{a} \times \frac{1}{f'(x)} \tan^{-1} \frac{f(x)}{a}$$

$$\begin{aligned} \text{b) } \int_0^{1/2} \frac{x}{1+4x^2} dx &= \frac{1}{8} \int_0^{1/2} \frac{8x}{1+4x^2} \\ &= \frac{1}{8} \left[\ln 1+4x^2 \right]_0^{1/2} \\ &= \frac{1}{8} (\ln 2 - \ln 1) \\ &= \frac{1}{8} \ln 2 \end{aligned}$$

$$\text{c) } y = 2 \sin^{-1}(x+1)$$

$$\begin{aligned} D: -1 \leq x+1 \leq 1 \\ -2 \leq x \leq 0 \end{aligned}$$



$$\begin{aligned} \text{d) } u &= e^x \\ du &= e^x dx \end{aligned}$$

$$\int_{-\ln 2}^0 \frac{e^x}{\sqrt{1-e^{2x}}} dx = \int_{\frac{1}{2}}^1 \frac{du}{\sqrt{1-u^2}}$$

$$x=0 \Rightarrow u=1$$

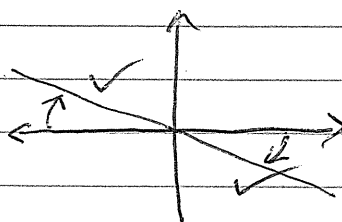
$$x=-\ln 2$$

$$= \ln \frac{1}{2} \quad u = e^{\ln \frac{1}{2}} = \frac{1}{2}$$

$$\begin{aligned} &= \left[\sin^{-1} u \right]_{\frac{1}{2}}^1 \\ &= \sin^{-1}(1) - \sin^{-1}\left(\frac{1}{2}\right) \\ &= \frac{\pi}{2} - \frac{\pi}{6} \\ &= \frac{\pi}{3} \end{aligned}$$

$$\text{e) } \tan x = -\sqrt{3}$$

$$x = n\pi - \frac{\pi}{3} \text{ for } n=0, \pm 1, \pm 2, \text{ or } n \in \mathbb{Z}$$



$$\text{f) } \lim_{\theta \rightarrow 0} \frac{\tan \theta}{2\theta} = \frac{1}{2} \lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = \frac{1}{2}$$

Question 7

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^{\frac{3\pi}{2}} (2 - \cos x)^2 dx \\
 &= \pi \int_0^{\frac{3\pi}{2}} 4 - 4\cos x + \cos^2 x dx \\
 &= \pi \int_0^{\frac{3\pi}{2}} 4 - 4\cos x + \frac{1}{2}(1 + \cos 2x) dx \\
 &= \pi \left[4x - 4\sin x + \frac{1}{2}\left(x + \frac{1}{2}\sin 2x\right) \right]_0^{\frac{3\pi}{2}} \\
 &= \pi \left[6\pi + 4 + \frac{3\pi}{4} + 0 - 0 \right] \\
 &= \frac{\pi}{4} (27\pi + 16) \quad \text{or equivalent} \quad \frac{27\pi^2}{4} + 4\pi
 \end{aligned}$$

$-\frac{1}{2}$ for small error at the end.

c) $f(x) = \frac{1}{\sqrt{4x+3}}$

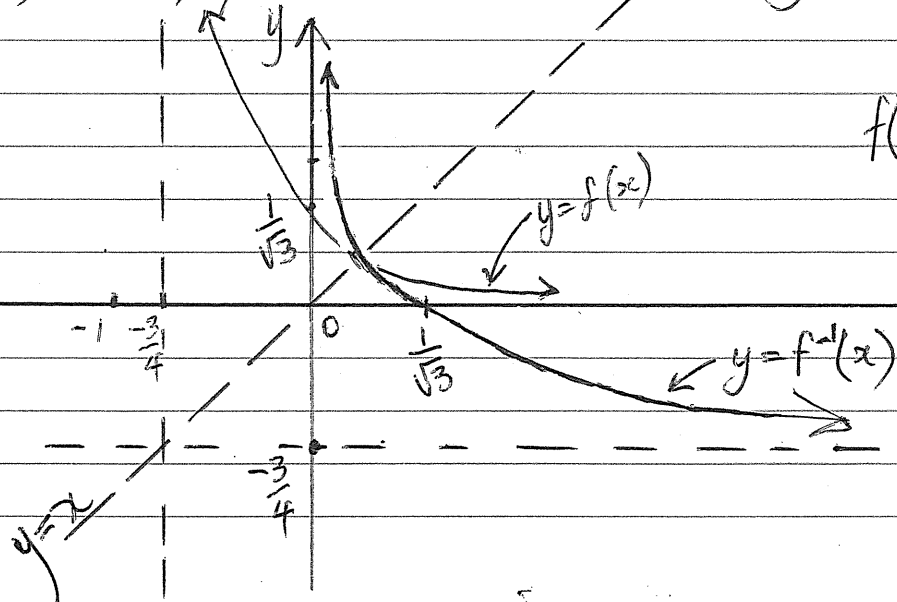
Vertical asymptote: $x = -\frac{3}{4}$

Horizontal asymptote: $y = 0$

(ii) $f'(x) = \frac{-1}{2}(4x+3)^{-3/2} \times 4$

$$= \frac{-2}{(4x+3)^{3/2}} \quad \text{or} \quad \frac{-2}{(4x+3)\sqrt{4x+3}}$$

(iii) Since $f'(x) < 0$, function is monotonically decreasing



$$f(0) = \frac{1}{\sqrt{3}} \approx 0.577$$

(iv) $y = f^{-1}(x)$ should have asymptotes $x=0$, $y = -\frac{3}{4}$

Two graphs meet on $y=x$

(v) Let $x = \frac{1}{\sqrt{4y+3}}$

$$x^2 = \frac{1}{4y+3}$$

$$\frac{1}{x^2} = 4y+3$$

$$\therefore y = \frac{1}{4} \left(\frac{1}{x^2} - 3 \right) \text{ or } y = \frac{1-3x^2}{4x^2}$$

Defined for $x > 0$ since range of $y = f(x)$ restricted to $y > 0$

Question 8

a) $y = \sin^{-1}(2x-1)$

$$y' = \frac{1}{\sqrt{1-(2x-1)^2}} \times 2$$

$$= \frac{2}{\sqrt{1-(2x-1)^2}}$$

$$\text{At } x = \frac{1}{4}, y' = \frac{2}{\sqrt{\frac{3}{4}}} = \frac{4}{\sqrt{3}}$$

$$\text{Equation of tangent: } y + \frac{\pi}{6} = \frac{4}{\sqrt{3}} \left(x - \frac{1}{4} \right)$$

$$y = \frac{4}{\sqrt{3}} x - \frac{1}{\sqrt{3}} - \frac{\pi}{6}$$

$$f(x) = 2(\sqrt{x}-1) - \log_e x, \quad x > 0$$

$$f'(x) = 2 \times \frac{1}{2} x^{-1/2} - \frac{1}{x}$$

$$= \frac{1}{\sqrt{x}} - \frac{1}{x}$$

$$= \frac{\sqrt{x}-1}{x}$$

(ii) Stat points occur when $f'(x) = 0$

$$\text{ie } \sqrt{x} = 1$$

$$x = 1$$

x	$\frac{1}{2}$	1	$1\frac{1}{2}$
$f'(x)$	-0.586	0	0.1498

There is a minimum at $(1, 0)$

(iii) $\lim_{x \rightarrow \infty} \frac{\sqrt{x}-1}{x} = 0$

Alternatively:

$$f''(x) = -\frac{1}{2} x^{-3/2} + \frac{1}{x^2}$$

$$= -\frac{1}{2x^{3/2}} + \frac{1}{x^2}$$

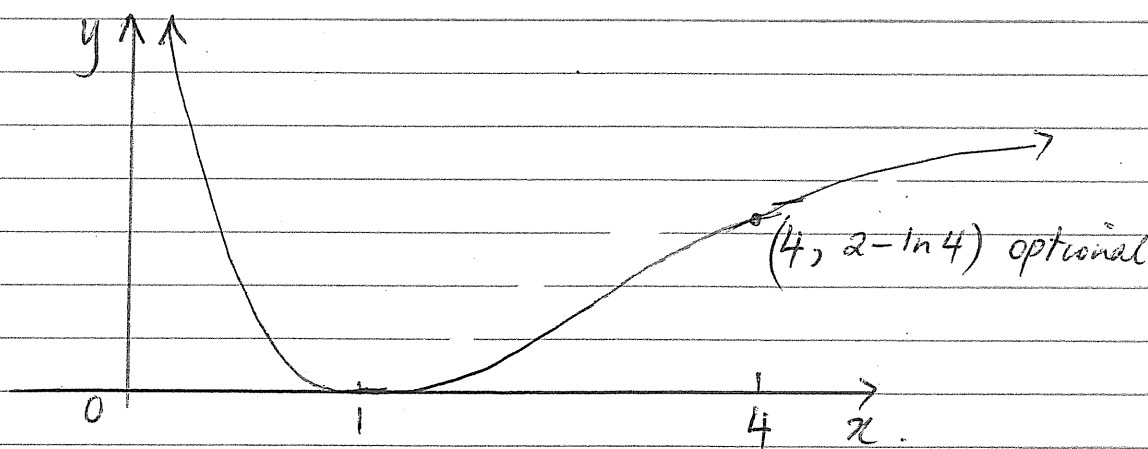
$$= \frac{-\sqrt{x} + 2}{2x^2}$$

$$= \frac{2-\sqrt{x}}{2x^2}$$

$$f''(1) = \frac{1}{2} > 0$$

$\therefore (1, 0)$ is a minimum

(iv)



(v) $2(\sqrt{x}-1) - \log_e x > 0$ from the graph, for $x > 1$

$$2(\sqrt{x}-1) > \log_e x$$

$$\frac{2}{x}(\sqrt{x}-1) > \frac{\log_e x}{x} \quad \text{since } x > 1 \text{ inequality holds true}$$

$$\text{ie } \frac{\log_e x}{x} < \frac{2}{x}(\sqrt{x}-1) \text{ for } x > 1$$